

TempoRF-DETR: Video-Based Coronary Stenosis Detection with Image-to-Video Distillation

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I. INTRODUCTION

Coronary artery disease (CAD) is one of the leading causes of mortality worldwide, driven by the build-up of atherosclerotic plaque that narrows (stenoses) the coronary arteries [1]. X-ray coronary angiography (XCA) is the clinical gold standard for assessing stenosis severity, but its interpretation is largely manual and exhibits substantial inter- and intra-observer variability [2][3], which directly affects subsequent treatment decisions. Automated stenosis detection in XCA is therefore of considerable clinical value, yet it remains challenging: due to fluoroscopic projection, vessels overlap and move with respiratory and cardiac motion. There is consequently a clear need for accurate and robust automated stenosis detection models.

Deep learning has shown considerable promise in this task. Early methods focused on single images or key frames, because frame-level annotation is simpler and image-based detection pipelines are easier to train and deploy. Representative examples include weakly supervised key-frame classification and heatmap localization [4], real-time object detectors operating on raw angiograms [5][6][7], and end-to-end single-frame pipelines that combine lesion detection with segmentation and quantification [8][9]. However, single-frame models make their decision from only one snapshot, even though angiography is acquired as a dynamic sequence: when overlap, poor contrast, or incomplete opacification appear in an individual frame, useful evidence is often present in adjacent frames. Recent work has therefore shifted toward video-based stenosis detection, where temporal aggregation consistently improves robustness and yields higher detection quality than frame-by-frame inference [10][11][12][13][14]. From an accuracy standpoint, a video-based model is the natural target architecture for a clinically useful detector.

Building such a model in the medical imaging domain, however, runs into two well-known obstacles. First, labeled medical data is expensive and scarce: pixel- or box-level annotation on angiograms has to be performed by trained cardiologists, which limits the size of any individual dataset [15]. Second, models suffer from domain shift: a detector trained on data from one institution or one type of X-ray system tends to lose substantial quality on data acquired from a different vendor or center, even when the underlying clinical task is identical [1][3]. A natural way to attack both problems at once is to train on several datasets simultaneously. This both increases the effective amount of supervision and, more importantly, forces the model to extract features that are common across acquisition setups rather than features tied to one particular scanner - encouraging cross-domain generalization. This effect has been empirically demonstrated in adjacent medical imaging modalities: pooled multi-site training improves segmentation accuracy on heterogeneous prostate MRI compared to any single-site baseline [16], and balanced multi-source sampling improves out-of-distribution generalization of chest X-ray pathology classifiers across ten datasets [17]. In short, multi-dataset training is an attractive route toward both data efficiency and robustness against domain shift.

Several public XCA resources are now available, and a defining feature of this landscape is that they come in two different modalities: some datasets are released as collections of individual angiography *images*, while others are released as full *video sequences*. As representative examples on either side of this split, ARCADE [18] provides region-annotated single-frame images, whereas Danilov et al. [5] released angiographic video series from one hundred patients acquired at the Research Institute for Complex Problems of Cardiovascular Diseases. Combining datasets across these two modalities is exactly what we want - it expands both the volume and the diversity of training data - but it is also what makes the problem non-trivial: a video-based detector cannot consume a static frame in the same way as a temporal sequence. Solving this *image-to-video data bridging* problem [19] - how to use 2D single-frame datasets when training a model whose target input is a video - is therefore central to our approach, on the same level as the design of the video detector itself.

The goal of this work is therefore twofold, and the two contributions are designed in symbiosis. First, we propose

TempoRF-DETR, an accurate and efficient video-based stenosis detector for XCA: a temporally-aware extension of RF-DETR that exchanges context across all frames of a clip in a single forward pass via an Early Temporal Fusion (ETF) module. Second, and equally central, we introduce an *image-to-video distillation* scheme to train TempoRF-DETR jointly on single-frame and video datasets, addressing labeled-data scarcity and cross-dataset domain shift. The intuition: video datasets can be decomposed into frames, so a 2D RF-DETR *teacher* can be trained on every available frame - single-frame datasets plus frames extracted from videos - and learns richer per-frame features than the video *student*, which only sees the smaller pool of temporally-annotated video. We therefore freeze the teacher and distill its per-frame knowledge into the student, leaving it free to learn temporal aggregation on top - letting single-frame datasets indirectly shape the video model even though they contain no video at all.

II. RELATED WORK

Single-frame stenosis detection. The dominant recipe in modern single-frame stenosis detection is to take a strong general-purpose object detector and apply it directly to raw angiograms. Essentially every major detector family has been instantiated this way. Two-stage detectors (Faster R-CNN, RFCN with different backbones) were benchmarked at real-time rates by Danilov et al. [5], who identified RFCN ResNet-101 V2 as the best speed-accuracy trade-off. Single-stage detectors are represented by EfficientDet, RetinaNet, and attention-augmented Faster R-CNN variants in Van Stralen et al. [6], and by the YOLO family (v5/v7/v8) in K peli et al. [7]. Beyond pure detection, several works pair a detector with a segmentation branch for quantitative coronary analysis (QCA): Tang et al. [8] combine SegFormer with an R-CNN lesion detector, and Choudhary et al. [9] integrate YOLO11m with DeepLabv3+ in their ODySSeI framework, adding a QCA-free severity estimation step. The common thread is that single-frame stenosis detection has effectively become a problem of *choosing* and *adapting* a powerful general-purpose detector for angiograms; the architectures themselves are not specific to medical imaging. All such methods, however, are fundamentally limited to a single snapshot and cannot exploit temporal context.

Video-based stenosis detection. A clear pattern runs through nearly all video-based stenosis detectors: they take a strong 2D image-level detector as the per-frame backbone and bolt on a *temporal aggregation module* that shares information across consecutive frames. Cong et al. [10] pair an Inception-v3 per-frame backbone with CNN+LSTM keyframe selection and an anchor-based FPN for localization, but struggle with small branch lesions. Pang et al. [11] use ResNet-50 + FPN as the per-frame backbone and add Sequence Feature Fusion and Sequence Consistency Alignment modules over 9-frame sequences; the alignment module is sensitive to vessel motion amplitude. Han et al. [12] keep the same ResNet-50 + FPN backbone and replace the temporal aggregator with a proposal-shifted Transformer that aligns vessel positions across time, at

a substantial computational cost. Li et al. [13] (STQD-Det) again build on ResNet-50 + FPN and add a Spatio-Temporal Feature Sharing module that Hungarian-matches detections across frames and projects features from confident frames onto ambiguous ones, with the box head reformulated as a quantum diffusion process. Li et al. [14] (LT-YOLO) inherit a YOLOv8 backbone enhanced with a dynamic Transformer block and add a Mamba-based bidirectional spatial-temporal fusion neck for long-term temporal mining, additionally producing categorical severity grading. The template is consistent across the field: the 2D backbone determines the per-frame feature quality, and the innovation happens in the temporal module on top—which is exactly the template TempoRF-DETR follows, reusing the powerful 2D RF-DETR detector as the per-frame backbone and adding our Early Temporal Fusion (ETF) module as the temporal aggregator. Two data-level limitations remain open across these works: most are validated on a single institution or vendor, leaving cross-dataset robustness underexplored [3], and public XCA resources are split between single-frame and video datasets [18][20], with most labeled data concentrated on the single-frame side—which is precisely the regime our image-to-video distillation scheme targets. Table I summarizes the reviewed methods.

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A. Abbreviations and Acronyms

Define abbreviations and acronyms the first time they are used in the text, even after they have been defined in the abstract. Abbreviations such as IEEE, SI, MKS, CGS, ac, dc, and rms do not have to be defined. Do not use abbreviations in the title or heads unless they are unavoidable.

TABLE I
SUMMARY OF REVIEWED SINGLE-FRAME AND VIDEO-BASED STENOSIS DETECTION METHODS.

Study	Task	Architecture	Input	Key Result
Stralen et al. [6]	Detection	EfficientDet (D3)	Single frame	mAP 0.67
Danilov et al. [5]	Detection	RFCN/Faster R-CNN	Single frame	mAP 0.94, 10 fps
Küpelı et al. [7]	Detection	YOLOv5/v7/v8	Single frame	8.3K-image benchmark
Tang et al. [8]	Det.+Seg.+QCA	SegFormer + R-CNN	Keyframes	End-to-end pipeline
Choudhary et al. [9]	Det.+Seg.+QCA	YOLO11m + DeepLabv3+	Single frame	Open-source, QCA-free
Cong et al. [10]	Class.+Det.	CNN+LSTM + FPN	Video	AUC 0.86 (patient)
Pang et al. [11]	Detection	ResNet-50 + FPN	9-frame seq.	F1 88.1%
Han et al. [12]	Detection	ResNet-50+FPN+Trans.	9-frame seq.	F1 79.11
Li et al. [13]	Detection	ResNet-50+FPN+QDM	9-frame seq.	STFS module
Li et al. [14]	Det.+Severity	LT-YOLO (Mamba)	Video seq.	mAP@0.5 76.4%

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$$a + b = \gamma \quad (1)$$

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TABLE II
TABLE TYPE STYLES

Table Head	Table Column Head		
	Table column subhead	Subhead	Subhead
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ACKNOWLEDGMENT

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